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IE 468- Pricing and Revenue Optimization
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NETWORK MANAGEMENT FOR NORSE ATLANTIC AIRWAYS

Norse Atlantic Airways is a small airline that operates comfortable 30-seat Embraer ERJ 135 aircrafts for exclusive service within Northern Norway. The company currently operates five services: three of which are non-stop flights. The following table summarizes the data for products (origin-destination-fares)

ODF	Origin	Destination	Connection	Class	Fare (NOK)	Demand
101	Bodø	Tromsø	Direct	M	2500	U(9,13)
102	Bodø	Tromsø	Direct	Y	1800	U(25,35)
103	Bodø	Alta	Tromsø	M	3500	U(7,11)
104	Bodø	Alta	Tromsø	Y	2000	U(13,19)
105	Tromsø	Alta	Direct	M	2200	U(7,9)
106	Tromsø	Alta	Direct	Y	1500	U(19,27)
107	Alta	Hasvig	Direct	M	2400	U(7,11)
108	Alta	Hasvig	Direct	Y	1200	U(13,29)
109	Tromsø	Hasvig	Alta	M	2000	U(7,13)
110	Tromsø	Hasvig	Alta	Y	1300	U(10,16)

The airline uses a network with three flights: WF 1001 (Bodø-Tromsø), WF 2002 (Tromsø-Alta) and WF 3003 (Alta-Hasvig). The notation $U(a,b)$ corresponds to discrete uniform distribution between (including) a and b . The company's goal is to determine how many seats should be allocated for each ODF on this network.

Question 1:

- A. Formulate this problem as a Deterministic Linear Program. Write down the formulation.
- B. Solve the program. Show the results.
- C. Suppose that there is a new customer from a partner airline who needs to go from Tromsø to Alta. What should be the minimum price that you would like to charge for this new request?
- D. Suppose that there is a new customer from a partner airline who needs to go from Bodø to Alta. What should be the minimum price that you would like to charge for this new request?

Question 2:

- A. Discuss why using the results of the Deterministic Linear Programming formulation may not be a very good idea.
- B. Norse Atlantic Airways wants to use virtual nesting to determine the booking limits for this flight. For each leg of the network, using the solution of the DLP, suggest virtual nests and assign products to these nests. Discuss how you can use these nests for booking control.

Question 3:

- A. Formulate this problem as a Probabilistic Non-Linear Program. Write down the formulation.
- B. Solve the program. Show the results.
- C. Suppose that there is a new customer from a partner airline who needs to go from Tromsø to Alta. What should be the minimum price that you would like to charge for this new request?
- D. Suppose that there is a new customer from a partner airline who needs to go from Bodø to Alta. What should be the minimum price that you would like to charge for this new request?

Question 4:

- A. Solve this problem using Randomized Linear Program. Create 20 random instances.
- B. Show the results.
- C. Suppose that there is a new customer from a partner airline who needs to go from Tromsø to Alta. What should be the minimum price that you would like to charge for this new request?
- D. Suppose that there is a new customer from a partner airline who needs to go from Bodø to Alta. What should be the minimum price that you would like to charge for this new request?